Department of Information Engineering Technology

UNEVOC Network Member

Advance Database Management System
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Lecture 8

Distributed Database Outline

- Introduction to Distributed Database
- Distributed DBMS
- Over-view of DDBS
- Replication
- Fragmentation
- Distributed catalog management
- Advantages and Disadvantges of DDBS

A distributed database (DDB)

- A distributed database (DDB) is combination of two contrasting technologies database system computer network.

- A distributed database (DDB) is a collection of multiple, logically interrelated databases distributed over a computer network.

Distributed Database Management system

- A distributed database management system (D–DBMS) is the software that manages the DDB and provides an access mechanism that makes this distribution transparent to the users.

Strategies of distribution

- Data replication
  - Copies of data distributed to different sites
- Horizontal partitioning
  - Different rows of a table distributed to different sites
- Vertical partitioning
  - Different columns of a table distributed to different sites
- Combinations of the above

Replication strategies

- Synchronous
  - All copies of the same data are always identical
  - Data updates are immediately applied to all copies throughout network
  - Good for data integrity
  - High overhead ➔ slow response times
- Asynchronous
  - Some data inconsistency is tolerated
  - Data update propagation is delayed
  - Lower data integrity
  - Less overhead ➔ faster response time

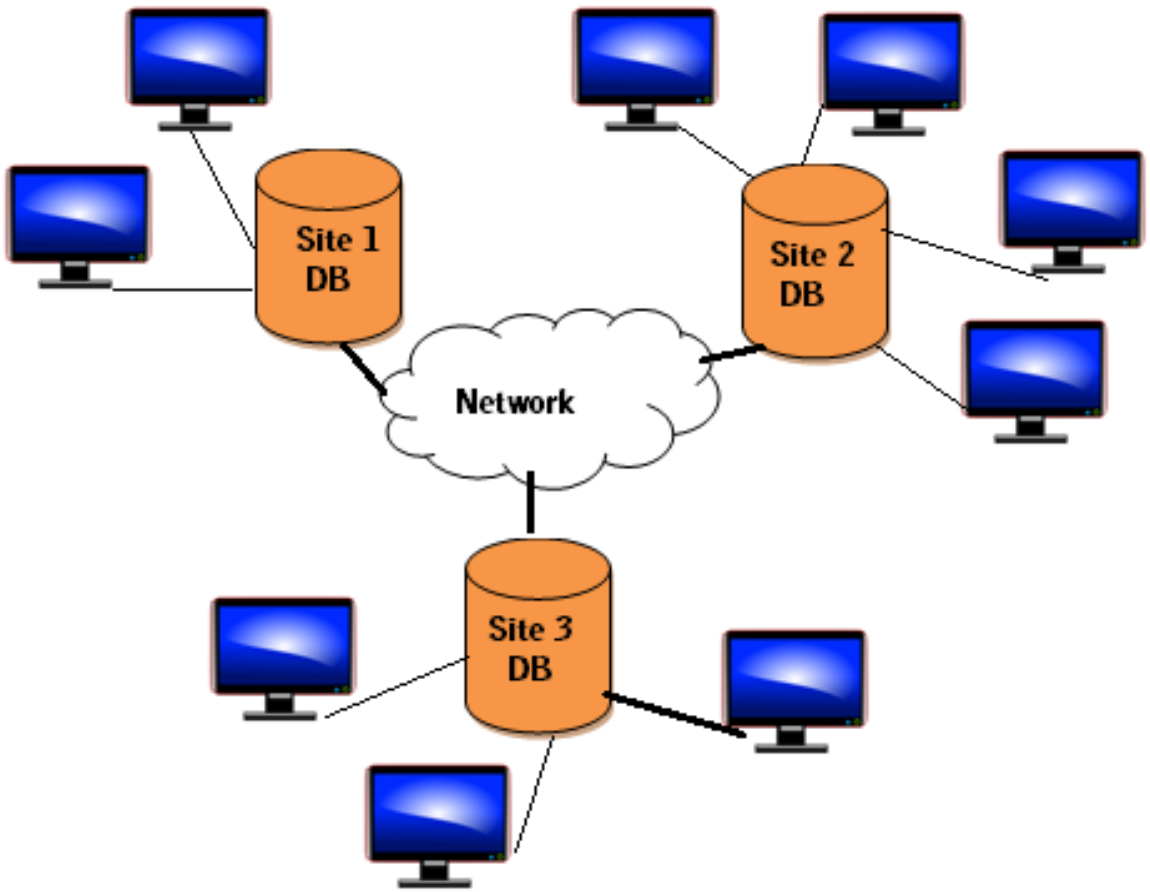
User Access in DDB

- User access database via application
- Two types of application
- Local
- Global
- Local application do not required data from remote site
- Global app required data from remote site

Key points

- Decentralized database system: A collection of independent databases on non-networked computers.
- Centralized database system: with network
- Distributed database system: distributed database (DDB) is a collection of logically interrelated databases over computer network.
- Distributed computing system: A number of autonomous processing elements that are connected through a computer network and that cooperate in performing their assigned tasks

Distributed DBS



Types of DDBS

- Homogenous DDBS
  - All sites have identical s/w
  - Are aware of each other
  - Appear to user as single database
- Heterogeneous DDBS
  - Use different schema and s/w
  - Schema difference is problem for query processing
  - S/w difference for transaction processing
  - May not aware of each other may provide limited facility.

Advantages

- Reduced Communication Overhead: Most data access is local, less expensive and performs better.
- Improved Processing Power: Instead of one server handling the full database, we now have a collection of machines handling the same database.
- Removal of Reliance on a Central Site : If a server fails, then the only part of the system that is affected is the relevant local site. The rest of the system remains functional and available.

Advantages conti…….

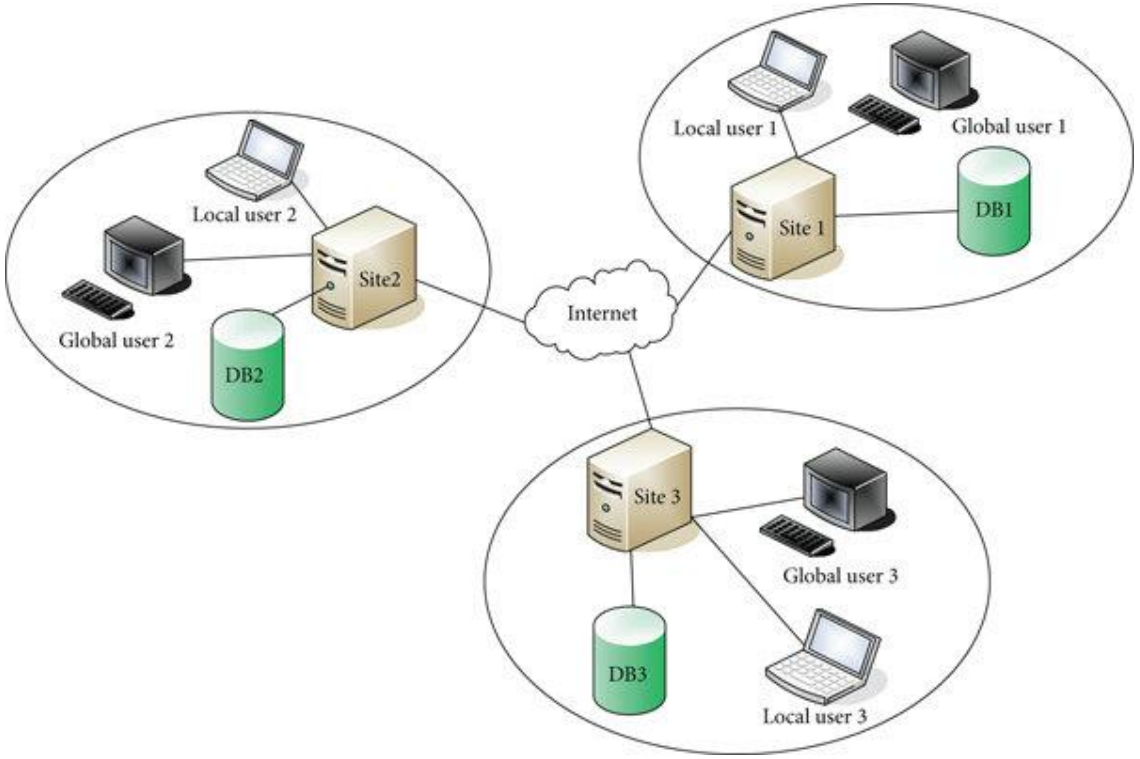
- Expandability: It is easier to accommodate increasing the size of the global (logical) database.
- Local autonomy: Local site can operate with its database when network connections fail . Each site controls its own data, security, logging, recovery.
- Location Transparency: User does not have to know the location of the data. Data requests automatically forwarded to appropriate site
- Backups,

Disadvantages

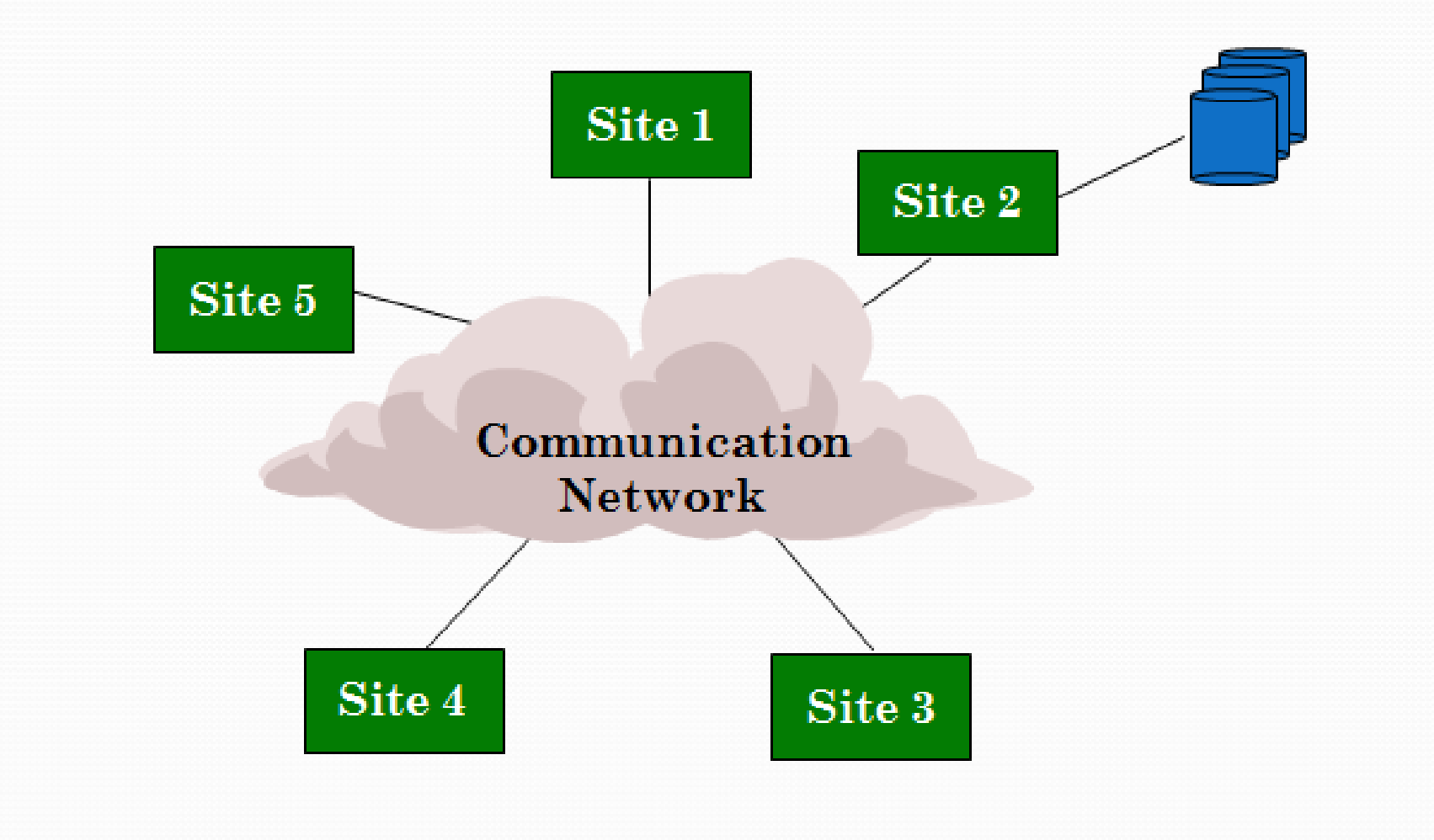
- Architectural complexity.
- Cost.
- Security.
- Integrity control more difficult.
- Lack of experience.
- Database design more complex.
- Software cost and complexity
- Slower response for certain queries

Distributed DBMS Vendors

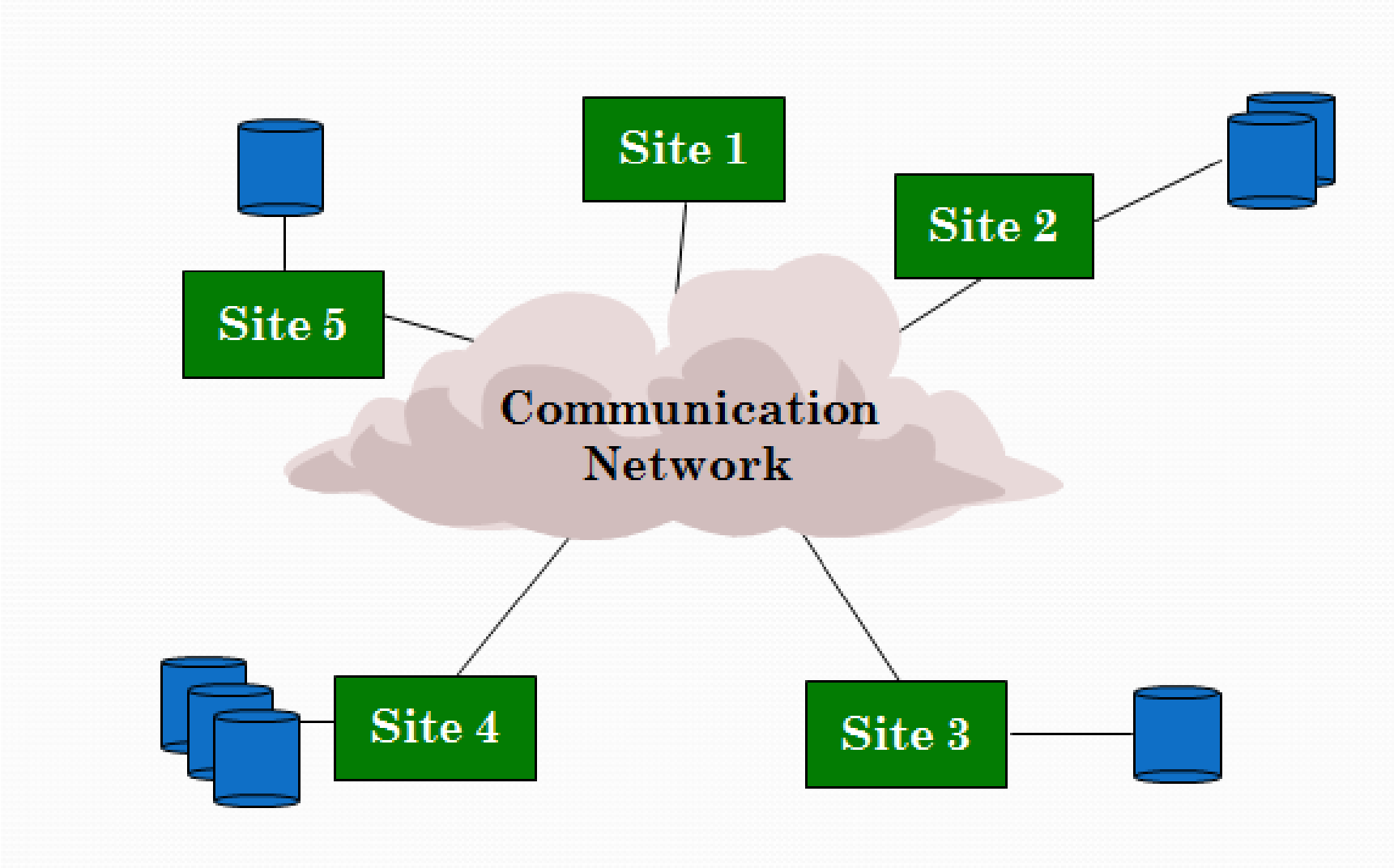
- Oracle
- Microsoft
- Informix
- Sybase
- IBM
- Computer Associates
- Ingress
- Others……



Centralized DBMS on a Network



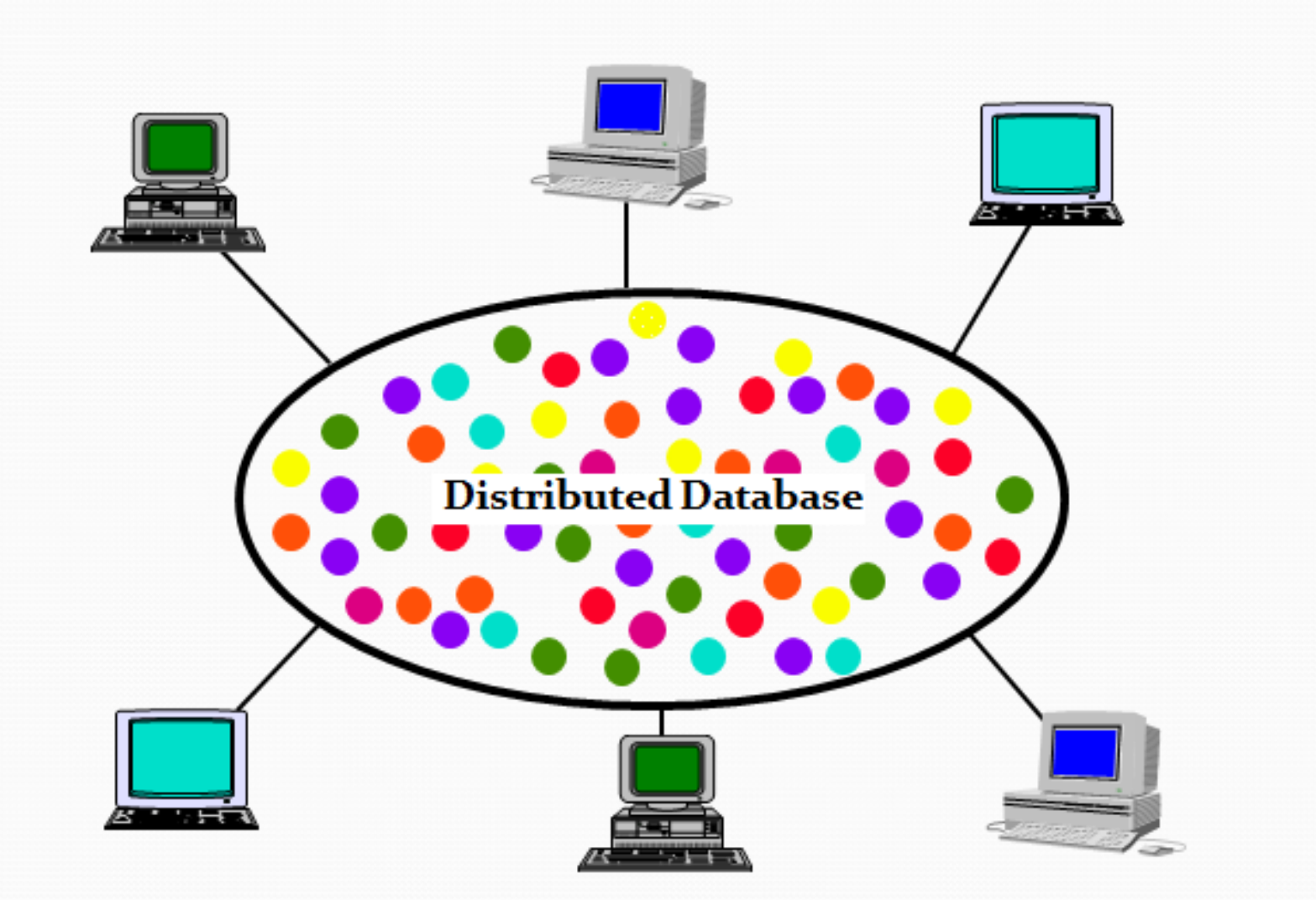
Distributed DBMS Environment



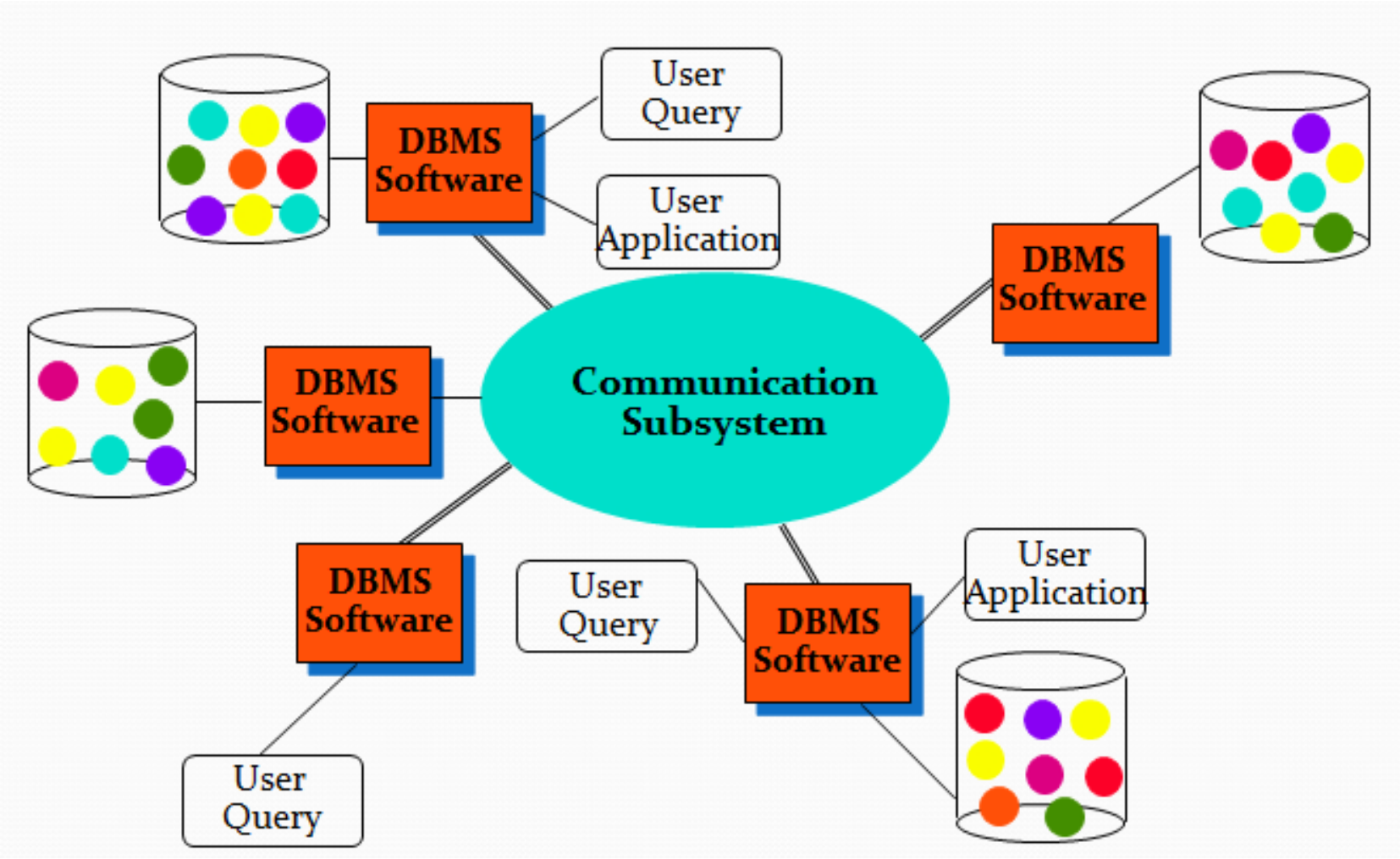
Transparency:

- Transparency is the separation of the higher level semantics of a system from the lower level implementation issues.
- A transparent system “hides” the implementation details from the users.
  - Network (distribution) transparency
  - Replication transparency
  - Fragmentation transparency
  - horizontal fragmentation:
  - vertical fragmentation:
  - Hybrid:

Distributed Database - User View



Distributed DBMS - Reality



Network (distribution) transparency

- Network/Distribution transparency allows a user to perceive a DDBS as a single, logical entity
- The user is protected from the operational details of the network (or even does not know about the existence of the network)
- The user does not need to know the location of data items and a command used to perform a task is independent from the location of the data and the site the task is performed (location transparency)
- A unique name is provided for each object in the database (naming transparency)

Network transparency conti……..

- Different ways to ensure naming transparency:
- Solution 1: Create a central name server; however, this results in
  - – loss of some local autonomy
  - – central site may become a bottleneck
  - – low availability (if the central site fails remaining sites cannot create new objects)
- Solution 2: Prefix object with identifier of site that created it
  - – e.g., branch created at site S1 might be named S1.BRANCH
  - – Also need to identify each fragment and its copies
  - – e.g., copy 2 of fragment 3 of Branch created at site S1 might be referred to as

Replication transparency

- Replication transparency ensures that the user is not involved in the management of copies of some data
- The user should even not be aware about the existence of replicas, rather should work as if there exists a single copy of the data
- Replication of data is needed for various reasons
- – e.g., increased efficiency for read-only data access

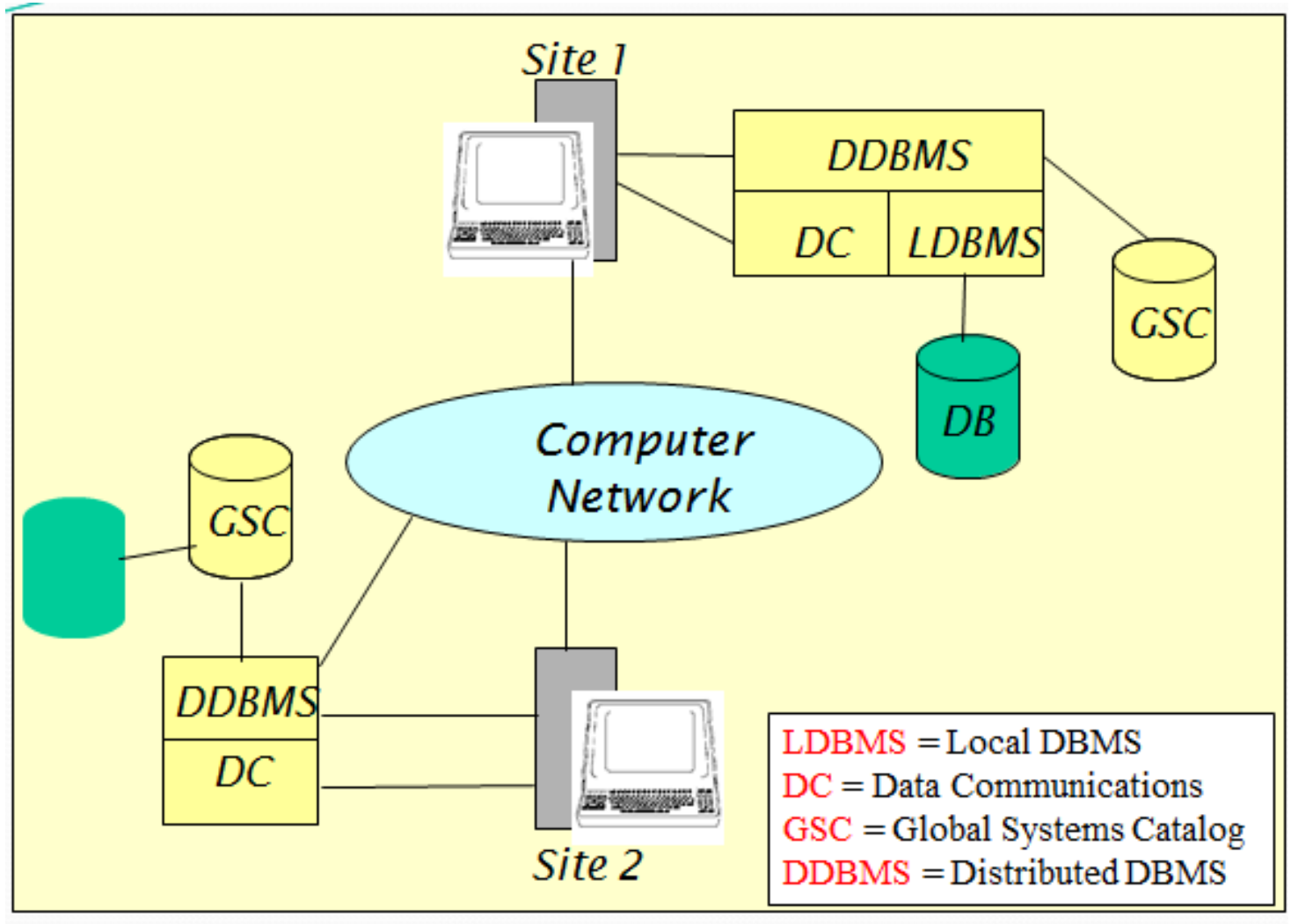
Fragmentation transparency

- Fragmentation transparency ensures that the user is not aware of and is not involved in the fragmentation of the data
- The user is not involved in finding query processing strategies over fragments or formulating queries over fragments
- The evaluation of a query that is specified over an entire relation but now has to be performed on top of the fragments requires an appropriate query evaluation strategy
- Fragmentation is commonly done for reasons of performance, availability, and reliability

Transaction transparency

- Transaction transparency ensures that all distributed transactions maintain integrity (assuring the accuracy) and consistency (A transaction is a correct transformation of the state) of the DDB and support concurrency.
- Each distributed transaction is divided into a number of sub-transactions (a sub-transaction for each site that has relevant data) that concurrently access data at different locations
- DDBMS must ensure the indivisibility of both the global transaction and each of the
- sub-transactions Can be further divided into
  - Concurrency transparency
  - Failure transparency

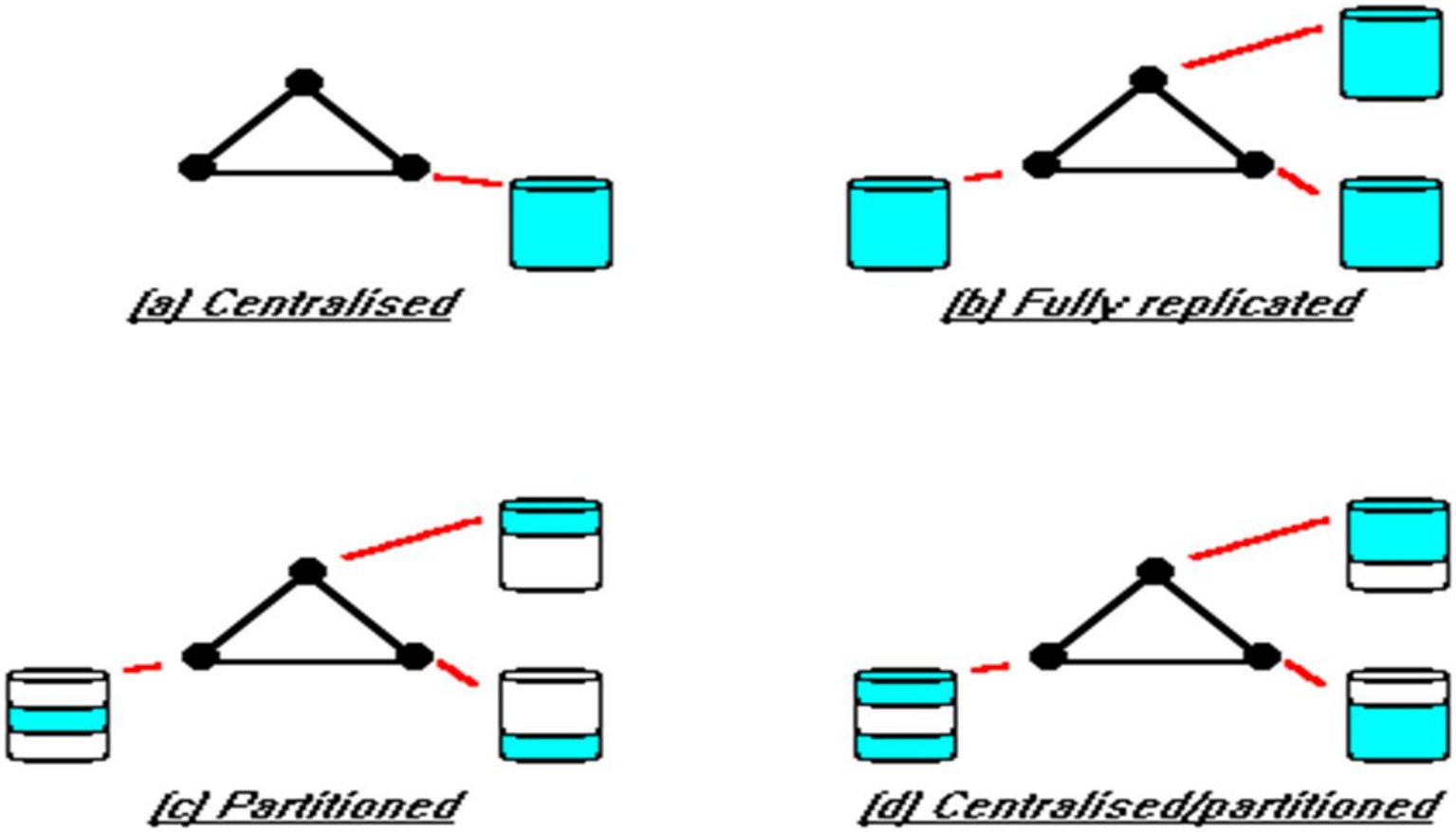
COMPONENTS OF A DDBMS



Distributed Catalog Management

- The distributed database catalog entries must specify site(s) at which data is being stored in addition to data in a system catalog in a centralized DBMS. Because of data partitioning and replication, this extra information is needed. There are a number of approaches.
- Centralized - Keep one master copy of the catalog
- Fully replicated - Keep one copy of the catalog at each site
- Partitioned - Partition and replicate the catalog as usage patterns demand
- Centralized/partitioned - Combination of the above

Catalog Management Approaches



Distributed DBMS Issues

- Concurrency Control
- Synchronization of concurrent accesses
- Consistency and Isolation of transactions' effects
- deadlock management. Deadlock management: prevention, avoidance, detection/recovery

Distributed Concurrency Control

- Concurrency control in distributed databases can be done in several ways. Locking and Time stamping are two techniques. but time stamping is generally preferred.
- The problems of concurrency control in a distributed DBMS are more severe because of the fact that data may be replicated and partitioned..

Time-stamping

- Time-stamping is a method of concurrency control where basically, all transactions are given a timestamp or unique date/time/site combination and the database management system uses one of a number of protocols to schedule transactions which require access to the same piece of data.
- While more complex to implement than locking, time-stamping does avoid deadlock occurring by avoiding it in the first place

Distributed query optimization

- In a distributed database the optimization of queries by the DBMS itself is critical to the efficient performance of the overall system. Query optimization must take into account the extra communication costs of moving data from site to site, but can use whatever replicated copies of data are closest, to execute a query.

- Distributed update propagation:

DATE'S TWELVE RULES FOR A DDBMS

A distributed system looks exactly like a non-distributed system to the user

- Local autonomy
- No reliance on a central site
- Continuous operation
- Location independence
- Fragmentation independence
- Replication independence
- Distributed transaction processing
- Hardware independence
- Operating system independence
- Network independence
- Database independence
- Distributed query independence

Any Question ??